Chanseung Lee

cslee20a@gmail.com

• Education

Kyung Hee University, Republic of Korea

- B.S. in Applied Physics & Electronic Engineering 2023 – 2027 (expected)

• Research Interests

Developing semiconductor-based photonic integrated circuits for next-generation quantum computers.

- Silicon Photonics: Towards integrated optical computers with strain-engineered on-chip lasers

- Quantum Photonics: Towards integrated quantum processors with strain-engineered 2D quantum devices

• Research Experience

Q-SPIN Laboratory (Quantum semiconductor Photonic Integrated Lab), KAIST
Advisor: Professor **Donguk Nam** Research Intern, Dec 2025 – Feb 2026

- Designing a **four single-photon source architecture** for **universal linear optics**, inspired by state-of-the-art integrated photonic quantum computing experiments

- Developing **SPDC-based multi-photon generation schemes** optimized for fiber-coupled, chip-compatible linear optical networks

• Projects

[1] Characterization and Parameter Extraction of Long- and Short-Channel MOSFETs (2025) [pdf]

Measured I–V characteristics using MS TECH probe station and CLARIUS software

Extracted Vt (ext/lin/sat), SS, Ioff, GIDL, DIBL, and gm

Analyzed long-channel vs short-channel behaviors and body-bias effects

Compared device scaling effects based on L = 10 μm and L = 0.35 μm transistors

[2] TCAD CMOS Process Simulation (2025) [pdf]

Built complete nmos fabrication flow: oxidation → implantation → annealing → metallization

Extracted electrical characteristics (ID–VG/VD, Vth, gm, ro)

Explored short-channel non-idealities such as DIBL and GIDL and high-k oxide effects

[3] Designing Multistage Analog Amplifier Using LTspice (2025) [pdf]

Designed differential → gain → output stages

Performed AC/DC/transient analysis and pole-zero estimation

Optimized gain, bandwidth, and power consumption

[4] Single Photon Verification and Photon Statistics Analysis Using HBT, SPDC, and GRA Experiments (2025) [pdf]

Measured normalized second-order correlation function $g^2(0)$ using HBT and GRA setups

Compared photon statistics of coherent, thermal, and SPDC-generated single-photon sources

Implemented SPDC photon pair generation using BBO crystal and verified photon-pair correlations

Demonstrated single-photon behavior through coincidence suppression (g^2 (0) → 0)

Analyzed effects of detector alignment, filtering, and classical vs quantum light sources

• Skills

- Simulation & Modeling

Silvaco TCAD (ATHENA, ATLAS)

LTspice (analog integrated circuit design)

- Programming

Python

- Device & Fabrication Knowledge

Semiconductor processes (CVD, PVD, ALD, plasma etching, lithography, wafer bonding, MEMS)

MOSFET/TFT/ LCD & OLED physics